Hardware Acceleration Pathways for Decentralized Energy-Based Models: A 2026 Strategic Analysis of the Carnot Architecture

Executive Summary and Architectural Context

The Carnot framework represents a fundamental paradigm shift in post-generation verification for Large Language Models (LLMs). By shifting the computational burden from deterministic token autoregression to probabilistic energy-landscape sampling, Carnot implements an Energy-Based Model (EBM) framework to verify and repair outputs. The architecture relies heavily on a block-Gibbs sampling infrastructure operating over a tiny-Ising substrate. The scaling parameters—ranging from $n = 128$ production spins to planned deployments of $n = 256+$, with $K = 100$ sweeps per inference request—demand a hardware environment fundamentally misaligned with the Single Instruction, Multiple Threads (SIMT) architecture of conventional GPUs.

The inference path evaluates a composite energy function defined as $E(y) = -\sum J_{ij} y_i y_j - \sum h_i y_i$, combined with k-AND verifier indicator sums ($k = 6$ to $k = 15$). Each accepted sample requires the evaluation of a soft-Gibbs residual rejection formula: $\mu^{\beta}_{res}(y) \propto \mu(y) \cdot \exp(-\beta V(y))$, where $V(y) = \sum \mathbf{1}\{y \notin \hat{S}_i\}$. Furthermore, Carnot imposes a rigorous sovereignty mandate. Hardware portability is treated as a foundational political and engineering requirement, ensuring that the entire inference and Phase 5 in-situ training stack—featuring adaptive-K Persistent Contrastive Divergence (PCD) with simulated-annealing—can operate entirely on open-source hardware paths without closed vendor lock-in.

This exhaustive analysis investigates the 2025-2026 hardware landscape, explicitly differentiating between architectures optimized for deterministic matrix-multiply-accumulate (MAC) operations and those capable of natively supporting stochastic sampling across non-convex Ising landscapes. The evaluation strictly applies Carnot's unyielding constraints: toolchain sovereignty (rule 1), distribution mirroring (rule 3), no vendor abstractions in the core repository (rule 7), and an absolute cost ceiling of $\leq \$10,000$ per deployment.

Substrate Viability Matrix for Carnot Architecture

Open / Yes   Hybrid / Simulated   Closed / No

Platform	Architecture Type	Toolchain Sovereignty	Native Gibbs Support	Hardware Cost Tier
Extropic Z1	Thermodynamic	Closed	Yes	Enterprise/Cloud
Cologne Chip GateMate A1	FPGA	Open	No	<$100
Tenstorrent Wormhole n150d	Custom ASIC / RISC-V	Open	No	~$1000
D-Wave Advantage	Quantum Annealer	Closed	Simulated	Enterprise/Cloud
Lightmatter Envise	Photonic	Closed	No	Enterprise/Cloud

Evaluation of near-term hardware substrates against Carnot deployment constraints. Native Gibbs sampling capability and fully open-source toolchains are critical gating requirements for Tier 1 sovereignty anchor deployment.

Data sources: Extropic, Cologne Chip, Tenstorrent, MDPI, Lightmatter

Section A: Currently-Available Hardware Ecosystem (2025-2026)

A.1 Thermodynamic Computing Architectures

Thermodynamic computing represents the theoretically optimal substrate for Carnot's block-Gibbs sampler. By intentionally abandoning the deterministic von Neumann abstraction, thermodynamic hardware harnesses intrinsic physical noise—such as thermal fluctuations in silicon—to perform stochastic operations directly. This eliminates the massive energy overhead associated with generating algorithmic pseudo-random numbers on digital logic gates.

Extropic Z1 and Development Ecosystem

Extropic has emerged as the principal commercial pioneer in this domain, designing Thermodynamic Sampling Units (TSUs) that sample directly from programmable probability distributions.[1] Rather than simulating probabilities through sequential arithmetic, the TSU effectively becomes the probability distribution, natively operating on the language of uncertainty required by generative models.[2]

The core architectural component is the probabilistic bit (p-bit), a circuit designed to fluctuate randomly between binary states based on controllable input voltages.[2] The architecture organizes these p-bits into bipartite grids, ensuring that each element communicates strictly with its immediate neighbors.[2] This localized topology is ideal for block-Gibbs sampling, as it permits the simultaneous updating of entire rows without data-movement bottlenecks, achieving an iteration time that remains constant regardless of the grid scale.[2] In the context of a fully parallelized $n = 128$ Ising model, a block-Gibbs update occurs in $\mathcal{O}(1)$ time, limited solely by the physical relaxation time of the analog circuitry.

The Extropic rollout strategy has been incremental. The X0 silicon prototype validated the core p-bit technology in Q1 2025.[2] This was followed by the XTR-0 in Q3 2025, an experimental platform connecting TSUs with traditional processors to permit algorithmic development.[2] The production-scale Extropic Z1 is scheduled for early access in 2026.[3] The Z1 is designed to be mass-manufacturable on standard CMOS processes and will feature hundreds of thousands of interconnected probabilistic circuits per chip, scaling into the millions across multi-card systems.[2]

While precise throughput numbers (samples/second) for the Z1 remain undisclosed pending benchmark embargoes, Extropic has claimed that generative AI workloads running on TSUs could realize energy efficiency improvements up to 10,000 times greater than comparable GPU architectures.[2] However, the sovereignty profile of the Extropic hardware is mixed. On the software front, Extropic has released THRML, an open-source Python library under the Apache-2.0 license, which allows for the simulation and definition of thermodynamic algorithms.[4] This satisfies the software abstraction requirement. Nevertheless, the physical silicon remains a proprietary, closed-stack vendor ASIC. While standard CMOS manufacturing suggests an eventual path to affordable small-buyer quantities, the initial 2026 access programs will likely be restricted to enterprise and elite academic partners, potentially exceeding the $\leq \$10,000$ hard budget for independent deployment.

Normal Computing CN101

Normal Computing, another prominent entity in the thermodynamic space, targets stochastic sampling for AI inference via its Thermodynamic Matrix Inversion architectures.[6] The company views deterministic CPUs as inherently limited for modern probabilistic AI workloads and leverages randomness and metastability to achieve high-throughput Bayesian inference.[6]

Normal Computing achieved tape-out of its first-generation physics-based ASIC, the CN101, which underwent successful data-production testing in August 2025.[6] Crucially for the Carnot framework, the CN101 prototype architecture incorporates four compute tiles, each harboring 64 state variables, interconnected via a reconfigurable Network-on-Chip (NoC).[9] This configuration supports 256-dimensional models [9], perfectly aligning with Carnot's Phase 3 deployment target of $n = 256+$.

Normal Computing's roadmap targets commercial releases in 2026 and 2028.[8] However, evaluating its fit against Carnot's constraints reveals significant gaps. The software stack details, particularly regarding PyTorch/JAX compatibility or open-source availability, are not publicly documented.[6] Without a verifiable open-source toolchain mirroring capability, the CN101 cannot function as a primary sovereignty anchor. Furthermore, there is no public indication that individual CN101 units will be available to small-budget buyers, as the company explicitly targets AI/HPC data centers.[7]

A.2 Analog and Quantum Ising Machines

Historically, researchers seeking to accelerate quadratic unconstrained binary optimization (QUBO) problems have turned to quantum annealers and coherent Ising machines. However, Carnot does not require pure optimization (finding the lowest-energy argmin); it requires high-fidelity *sampling* from a Boltzmann distribution defined by the specified EBM.

D-Wave Advantage 2 Series

D-Wave's quantum annealers are the most widely recognized commercial platforms for Ising model evaluation. The Advantage system features over 5000 logical qubits.[10] However, repurposing a quantum annealer for classical Boltzmann sampling introduces severe theoretical and practical complications.

Extensive independent benchmarking of the D-Wave 2000Q system on Quantum Boltzmann Machine (QBM) image synthesis tasks demonstrates that D-Wave sampling does not significantly outperform classical generative models like Restricted Boltzmann Machines (RBMs) or Markov Chain Monte Carlo (MCMC) simulations evaluated via Fréchet Inception Distance (FID) and Kernel Inception Distance (KID).[11] When an annealer attempts to sample a distribution, it suffers from a phenomenon known as "freeze-out." This occurs when the quantum state stops evolving dynamically long before the annealing schedule concludes, locking the system into a distribution that is inherently non-Boltzmann and heavily skewed toward specific local minima.[12] Independent analyses comparing Gibbs sampling against D-Wave sampling reveal that the two techniques produce overlapping but unfavorably constrained probability valleys.[12]

Furthermore, the physical topology of the Advantage system (utilizing Pegasus or Zephyr connection graphs) limits direct variable interactions.[10] Mapping a fully-connected $n = 128$ Carnot EBM requires complex minor-embedding, wherein multiple physical qubits represent a single logical variable.[10] This drastically reduces the effective variable count, introduces chain-break errors, and injects embedding overhead that nullifies any quantum speedup during the sampling process. Finally, D-Wave hardware is exclusively accessible via cloud APIs, with sustained sampling easily violating the $\$500$ /month cloud cost ceiling. The architecture represents a profoundly closed, centralized system, completely disqualifying it from sovereignty consideration.

Photonic, Coherent, and Magnetic-Oscillator Ising Machines

The landscape of alternative analog Ising machines—including Coherent Ising Machines (CIMs) developed by NTT, Memcomputing Inc.'s self-organizing logic gates, spatial photonic Ising machines based on spatial light modulators (SLMs), and magnetic-oscillator implementations—remains fundamentally confined to academic research or highly guarded enterprise silos.

Extensive literature review reveals no commercial accessibility for small-buyer physical deployment in 2026. Companies like Memcomputing Inc. have historically focused on software-emulated ASICs for deterministic optimization routing, not thermal equilibrium sampling. The spatial photonic platforms documented in major literature (e.g., Pierangeli et al.) are tabletop optical setups requiring laboratory calibration, offering no ruggedized PCIe integration path. The absence of searchable commercial paths for these platforms acts as a definitive signal of vaporware or non-disclosure agreement (NDA)-exclusive access, rendering them entirely incompatible with Carnot's decentralized deployment ethos.

A.3 Photonic Compute Accelerators

Photonic computing exploits the physical properties of light—speed, parallelism, and wavelength multiplexing—to replace electronic multiply-accumulate (MAC) operations.[13] The industry is led by Lightmatter, alongside competitors like Lightelligence, Q.ANT, and Salience Labs.[14]

Lightmatter has successfully advanced the Envise photonic AI accelerator and the Passage interconnect platform.[15] Envise utilizes four photonic chips manipulating 512 light beams through over 200,000 optical components to perform matrix multiplication natively via optical interference, achieving a theoretical efficiency of 262 Tera-Operations per Watt (TOPS/W).[15] Passage acts as an advanced 3D-integrated photonic circuit providing immense bandwidth between dies, aiming for near-packaged optics by 2026 and co-packaged optics by 2027.[15]

Despite these impressive specifications, photonic platforms are structurally incompatible with Carnot’s core loop. Envise and its counterparts (e.g., Lightelligence's optical PIC switching) [17] are strictly designed for deterministic linear algebra, specifically accelerating dense deep neural networks like ResNet and BERT.[15] The underlying physics of Mach-Zehnder interferometric networks or incoherent intensity-modulation crossbar arrays cannot natively compute stochastic bit-flips or implement localized block-Gibbs sampling.[13] Furthermore, these platforms are deeply embedded within hyperscale data center infrastructure, strictly closed-source, and far exceed the small-lab budget constraints.[15] They offer zero utility for Carnot's Boltzmann sampling workloads.

A.4 Open-FPGA Pathways

Field Programmable Gate Arrays (FPGAs) represent the purest manifestation of Carnot's sovereignty mandate. By utilizing hardware capable of being synthesized, routed, and programmed entirely via open-source tools without vendor licensing, Carnot can achieve complete decentralization.

Cologne Chip GateMate A1 and A25

The GateMate series by Cologne Chip has established a benchmark for low-cost, high-efficiency, open-source-friendly FPGAs. Manufactured on a 28nm Super Low Power (SLP) process, the A1 variant features 20,480 8-input logic elements (CPEs) and operates with an ultra-low leakage current of 12 mA at 1.0V.[18]

The primary advantage of the GateMate is its native support for a completely open-source RTL-to-bitstream flow. Synthesis is executed via Yosys, place-and-route is managed by nextpnr (utilizing the prj-peppercorn architecture definition), and bitstream flashing is handled by openFPGALoader.[18] The pricing is highly disruptive: small quantities of evaluation boards and modules range from approximately $\$60$ to $\$100$ [19], rendering it universally accessible.

For Carnot, an $n = 128$ quadratic Ising solver fits comfortably within the 20,480 logic elements, provided the internal multiplier paths are optimized. However, the evaluation of the $k = 15$ verifier (the soft-Gibbs residual rejection) requires substantial logic for indicator sums. The GateMate A1 possesses only 1.28 Mb of Block RAM (BRAM).[18] This constrained memory space demands that the EBM verifier weights be heavily pipelined to prevent memory-fetch bottlenecks during the $K = 100$ sweeps per inference cycle. Furthermore, a known limitation of the open-source Yosys/Nextpnr stack is its historical deficiency in handling rigorous, highly complex timing constraints natively during synthesis.[20] Closing timing on high-frequency stochastic routing across the 8-input LUT trees will require extensive manual intervention in the Verilog generation scripts.

Lattice ECP5 and GOWIN GW5A

Lattice ECP5 architectures possess robust backing from the open-source community through Project Trellis and Yosys. ECP5 development boards, such as the Colorlight series, are inexpensive and ubiquitous.[21] The higher-end ECP5 variants contain significantly larger BRAM and DSP resources than the GateMate A1, providing sufficient capacity to house both the pseudo-random number generators (e.g., XORSHIFT arrays) necessary for digital Gibbs sampling and the complex combinatorial logic of the k-AND verifiers.

The Gowin GW5A series offers excellent cost-to-logic ratios and is supported by Project Apicula.[23] However, the open-source bitstream generation for Gowin relies heavily on reverse-engineering and fuzzing the vendor bitstreams.[23] Consequently, the open toolchain frequently lags behind the silicon releases. If a specific DSP slice required for the verifier evaluation remains unsupported by Apicula, it introduces a critical sovereignty risk, potentially forcing a fallback to proprietary vendor software.

Microchip PolarFire SoC

The PolarFire SoC Discovery Kit presents a unique and compelling architectural hybrid. It combines a quad-core, RISC-V application-class processor capable of running Linux directly alongside 95K low-power FPGA logic elements.[24] Retailing for under $\$200$, it meets all cost constraints.

In the Carnot architecture, the RISC-V cores can manage the high-level Python orchestrations, execute the JAX/PyTorch verifier composition, and handle the soft-Gibbs residual rejections.

Simultaneously, the adjacent FPGA fabric can serve as the localized $n = 128$ block-Gibbs tiny-Ising substrate. This configuration drastically reduces the PCIe data-transfer bottlenecks inherent in discrete GPU setups. The primary obstacle is the toolchain; Microchip's Libero software is proprietary and licensing-based.[25] While Yosys support for PolarFire exists, it lacks the maturity of the ECP5 ecosystem, requiring significant upstream engineering to achieve full Rule 1 compliance.

Open-FPGA Platform	**Logic Elements**	**Open Toolchain Support**	**Cost ($)**	**BRAM Capacity**	**Sovereignty Risk**
Cologne Chip GateMate A1	20,480	Yosys, Nextpnr, openFPGALoader	~$60-$100	1.28 Mb	Very Low (Native OSS)
Lattice ECP5	Up to 84,000	Yosys, Project Trellis	~$50-$150	Up to 3.7 Mb	Low (Mature reverse-eng)
GOWIN GW5A	Up to 138,000	Project Apicula (Fuzzing)	~$100-$200	Variable	Medium (Fuzzing lag)
Microchip PolarFire SoC	95,000 + Quad RISC-V	Partial Yosys, Closed Libero	~$130	2.9 Mb	High (Closed vendor tools)

A.5 Custom ML ASICs (Spatial Architectures)

When FPGAs lack the fabric density for $n = 256+$ production scaling, but FPGAs' open nature is desired, specific spatial ML ASICs provide a vital bridge.

Tenstorrent Wormhole and Blackhole

Tenstorrent provides a stark philosophical counterpoint to Nvidia's closed CUDA ecosystem by architecting spatial computing platforms built on RISC-V.[26] The Wormhole architecture comprises a 2D grid of compute elements ("Tensix cores") interconnected via a high-bandwidth Network-on-Chip (NoC) and buffered by local SRAM.[29] A 2026 academic study utilizing the Wormhole architecture for sparse numerical algorithms (conjugate gradient solvers) demonstrated that while a single Tensix die operates at a fraction of the speed of an Nvidia H100 GPU for maximal matrix sizes, its performance relative to thermal design power (TDP) and hardware cost is highly favorable.[29]

For Carnot, the NoC and local core architecture is exceptional. An $n = 128$ Ising graph can be spatially partitioned across the Tensix cores. Unlike traditional GPUs, which suffer severe SIMT warp divergence when executing stochastic branching (a prerequisite for thermal sampling), the independent RISC-V controllers on each Tensix core can execute localized Markov chains or parallel block-Gibbs sweeps without stalling adjacent threads.

The Wormhole PCIe cards are priced aggressively for small buyers: the single-processor n150d retails at $\$1,099$ (160W), and the dual-processor n300d retails at $\$1,449$ (300W).[28] Most importantly, Tenstorrent satisfies Rule 3 (Distribution mirroring). The hardware is supported by TT-Metalium, a low-level, fully open-source SDK that allows developers direct bare-metal access to program the NoC routing and Tensix execution units.[32] This open software capability makes Tenstorrent the premier sovereignty-compliant ASIC on the market.

Cerebras, Groq, and SambaNova

Conversely, Cerebras (CS-3), Groq, and SambaNova prioritize massively dense, deterministic linear algebra. The Groq LPU (Language Processing Unit) is optimized entirely for ultra-low-latency deterministic token generation, functioning as the antithesis of the thermal-relaxation compute paradigm required for efficient block-Gibbs sampling. Cerebras offers unparalleled wafer-scale integration but remains securely locked behind enterprise cloud APIs and massive capital expenditures, fundamentally violating the cost ceiling, sovereignty constraint, and decentralization requirements.

A.6 Consumer Edge NPUs (The Sovereignty Anchor)

Rule 1 specifies that closed-stack vendor hardware can serve as a bonus tier, ensuring Carnot operates on hardware users already possess. The proliferation of Neural Processing Units

(NPUs) in consumer edge devices necessitates evaluation.

- **AMD Ryzen AI / XDNA (Strix Point):** Integrated APUs with up to 67 GB of unified memory [35] provide significant bandwidth. AMD has heavily invested in the ROCm and Ryzen AI 1.x SDKs, maturing support for ONNX and direct PyTorch execution. While highly efficient for static MAC graphs, generating localized randomness for Gibbs sampling forces NPU pipeline stalls. However, they remain excellent fallback hardware for executing the deterministic k-AND verifiers.
- **Intel AI Boost (Lunar Lake):** Intel's OpenVINO toolkit is robust, allowing heterogeneous execution. EBM workloads can be partitioned, pushing dense matrix computations to the integrated GPU while routing sparse verifier checks to the NPU.
- **Qualcomm Hexagon & Apple Neural Engine:** The Qualcomm Neural Network (QNN) SDK remains notoriously opaque on Linux platforms, violating the vendor mirror constraint unless strictly relegated to an ONNX fallback path. Similarly, Apple's CoreML abstracts the Neural Engine completely away from bare-metal manipulation. While necessary for ubiquitous deployment, neither architecture allows for the adaptive-K Persistent Contrastive Divergence algorithm to run optimally.

Section B: Coming Hardware (12-18 Month Horizon)

The hardware landscape from late 2026 into early 2028 indicates a shift toward specialized non-convex optimization and expanded open-source hardware design.

2027 Thermodynamic Computing Roadmaps

Normal Computing has explicitly stated a 2028 roadmap scaling to larger architectures designed for complex video and photo diffusion models.[8] Extropic's roadmap implies that the 2026 Z1 will be followed by denser physical integration of p-bits.[2] If these paradigms succeed in standardizing integration layers beneath PyTorch/JAX ecosystems, they will render pseudo-random digital MCMC sampling practically obsolete for generative tasks.

Open-Source ASIC Tape-outs (OpenROAD)

The OpenROAD project is aggressively pursuing "no-human-in-loop, 24-hour design" environments to automate SoC layout generation without power, performance, or area (PPA) degradation.[36] As of 2025, tools utilizing GenAI to optimize open-source layout synthesis are maturing.[36] Within the 18-month horizon, it is increasingly feasible for decentralized, open-source communities to tape out custom $n = 256$ Ising ASICs specifically tuned for the Carnot framework via accessible foundries like Efabless. A community-designed ASIC, defined by open standard cell libraries and fully reproducible GDSII files, would perfectly fulfill the sovereignty mandate.

1-bit, Ternary Bit Networks (TBN), and MTJ Hardware

The search for highly efficient stochasticity has led academic researchers to explore Magnetic Tunnel Junctions (MTJs). By exploiting the inherent thermal instability of low-barrier nanomagnets, MTJ p-bits act as true random number generators operating at nanosecond scales, evaluating Bayesian networks orders of magnitude faster than CMOS-based pseudo-random implementations. While the hardware startup ecosystem is aggressively expanding in this sector [37], transitioning this material science from academic laboratories (e.g., Purdue University studies) to a commercial, Linux-compatible PCIe form factor accessible to a small lab budget before late 2027 remains highly improbable. Similarly, commercial hardware natively restricted to 1-bit or Ternary logic exclusively for neural execution remains absent from the accessible consumer market.

Section C: Software Stacks and Abstraction Interfacing

Carnot's Rule 7 requires that all hardware-specific execution remain isolated beneath an abstract SamplerBackend protocol within the core repository. The open-source layer bridging higher-level JAX/PyTorch commands to the silicon dictates the success of any platform.

Topological Flexibility and Arbitrary Ising Graphs

A functional EBM software stack must support arbitrary Ising graph topologies, including highly connected structures with signed negative and positive couplings (J_{ij}). Hardware abstractions that force graphs into strict physical topologies (such as the Pegasus lattice required by D-Wave) [10] necessitate intermediate minor-embedding software. This mapping introduces catastrophic computational overhead and invalidates the speed of the underlying hardware. Extropic's THRML library provides a robust template, permitting the programmatic definition of energy functions and interactions regardless of logical constraints.[1] Tenstorrent's TT-Metalium similarly allows software to dynamically map logical topologies onto the physical NoC fabric bridging the Tensix cores.[30]

Block-Gibbs vs. Single-Site Metropolis-Hastings

Hardware abstractions must not strictly enforce single-site Metropolis-Hastings (MH) execution. In MH, variables are updated sequentially to satisfy the strict detailed balance requirement, which severely restricts throughput. Extropic circumvents this limitation theoretically by implementing bipartite grid partitioning, which allows the hardware to simultaneously update entire non-interacting rows.[2]

Custom FPGA and Tenstorrent implementations must be capable of replicating this chromatic grouping at the software abstraction layer. FPGAs excel here: an open-source Verilog design synthesized via Yosys can be explicitly programmed to evaluate an entire bipartite set of 64 spins in a single clock cycle. NPUs and SIMT GPUs, bound by rigid MAC vector lengths and branch-prediction algorithms, struggle significantly to emulate this simultaneous stochastic update without incurring massive memory-coalescing penalties.

Exposing Sample Distributions and Bitstream Reproducibility

The SamplerBackend must expose the raw probability distributions generated by the hardware, not merely the optimized argmin state. For the open-FPGA tier, reproducing the hardware interface is critical. The Carnot repository must host the parameterized Verilog, Amaranth, or LiteX generation scripts required to synthesize the $E(y)$ evaluation logic into a bitstream directly on the user's local machine. Relying on pre-compiled, opaque binary bitstreams violates the open-source mandate. The Yosys/Nextpnr toolchain ensures that any third party can verify, recompile, and flash the specific FPGA routing directly from the source code.[18]

Section D: Strategic Recommendations

Based on the exhaustive synthesis of the 2025-2026 computing ecosystem, the following strategic deployments are advised for the Carnot EBM framework.

1. Top 3 Platforms by Sovereignty Profile (≤$10k, Open Toolchain)

1. **Cologne Chip GateMate A1 / A25 (FPGA):** Priced at approximately $\$60 - \$100$ per evaluation module [19], the GateMate provides an uncompromised 100% open-source toolchain (Yosys, Nextpnr, openFPGALoader).[18] It acts as the ultimate cost-effective sovereignty anchor, capable of locally synthesizing discrete $n \leq 128$ Ising solvers without any proprietary licensing barriers.
2. **Tenstorrent Wormhole n150d (Spatial ASIC):** Retailing at $\$1,099$ [32], this is the most powerful platform that thoroughly satisfies the sovereignty mandate. The TT-Metalium stack grants complete bare-metal access to the RISC-V array and the Tensix compute cores [33], allowing Carnot engineers to construct customized parallel block-Gibbs samplers free from Nvidia CUDA lock-in.
3. **Lattice ECP5 FPGAs (via Colorlight Boards):** Featuring wide commercial availability and mature open-source integration via Project Trellis [21], the larger BRAM reserves on ECP5 boards offer a safer, more robust fallback for open-source digital Gibbs sampling than the fuzzing-reliant Gowin ecosystem.

2. Top 3 Platforms by Raw Sample-Throughput (n=128 Ising)

1. **Extropic Z1 (Thermodynamic ASIC):** By discarding algorithmic pseudo-randomness and utilizing physical p-bits governed by native hardware thermal relaxation [1], the production Z1 will outperform any deterministic digital hardware by several orders of magnitude in both samples/second throughput and sample/watt efficiency.[3]
2. **Dual Nvidia RTX 4090/5090 (CUDA/JAX):** While philosophically opposed to the

project's decentralized, vendor-agnostic ethos, top-tier consumer GPUs running heavily optimized Triton kernels or pure JAX vmap logic remain the brutalist benchmark for raw computational bandwidth. They can brute-force massive parallel MCMC chains despite inherent SIMT warp divergence.
3. **Tenstorrent Wormhole n300d:** Operating with dual processors at 300W [33], the spatial logic mapping across the NoC permits rapid, low-latency communication [29], significantly outperforming equivalent GPUs in complex, non-batched sequential sampling.

3. Top 3 NPU Paths for Verifier-Edge Inference

1. **AMD Ryzen AI / XDNA (Strix Point):** AMD's significant investments in open-source Linux drivers provide the lowest-friction pathway for local verifier offloading via the unified ROCm/XDNA stack.
2. **Intel AI Boost (Lunar Lake):** The maturity of OpenVINO in managing heterogeneous execution enables Carnot to execute heavy LLM token generation on the GPU while routing the k-AND verifier evaluation logic exclusively to the highly efficient NPU.
3. **Apple CoreML (Neural Engine):** Despite existing as a completely closed hardware ecosystem, the sheer ubiquity of Apple Silicon makes ANE support practically mandatory for generalized end-user deployment. EBM verifiers can be exported as statically shaped ONNX models and consumed by CoreML, though with severe limits on low-level diagnostic feedback.

4. Contrarian Pick: Microchip PolarFire SoC Discovery Kit

While the industry fixates on scaling massive PCIe accelerators, the $\$130$ Microchip PolarFire SoC [24] represents an optimal, overlooked paradigm for the Carnot framework.

Justification: The adaptive-K Persistent Contrastive Divergence process requires continuous dynamic supervision of the sampling temperature and the mathematical evaluation of the soft-Gibbs residual rejection. Managing this via PCIe bus data transfers from a host CPU to an external FPGA injects fatal latency bottlenecks into the $K = 100$ inference loop. The PolarFire SoC integrates a quad-core Linux-capable RISC-V processor on the exact same die as a 95K-element FPGA fabric.[24] This allows the RISC-V application cores to calculate the complex mathematical residuals natively, while employing the adjacent FPGA logic strictly as an ultra-fast, zero-latency co-processor for the $n = 128$ block-Gibbs spin updates. It completely eliminates the PCIe bottleneck, providing a sovereign, ultra-low-power, physically integrated node.

5. Red-Team Critique of the Current Carnot Portfolio

A rigorous, hardware-savvy review of Carnot's existing hardware array (RTX 3090 + Strix Point +

KV260 + future Z1) reveals critical architectural vulnerabilities, primarily regarding **memory hierarchy constraints and analog drift**.

The Weakest Claim (Strix Point APU Memory Integration): Relying on AMD Strix Point unified memory to accelerate tight EBM feedback loops is technically flawed. While the APU boasts 67 GB of unified memory [35], sharing physical memory between the CPU and the integrated GPU does not magically resolve cache coherence overheads or synchronization delays. Block-Gibbs sampling relies entirely on ultra-fast, tiny local memory (SRAM), not high-bandwidth GDDR or unified LPDDR channels. Pumping rapid state updates across the APU's internal fabric for $K = 100$ continuous sweeps per generated LLM token will almost certainly saturate the unified memory controller, simultaneously starving the primary LLM generation process and destroying inference throughput.

The Hidden Technical Debt of the KV260: Using the Xilinx KV260 as a proof-of-concept board establishes severe technical debt. The AMD/Xilinx Vivado toolchain is a completely closed-source, monolithic suite. Porting Vivado-centric HDL structures to the fully open-source Yosys/GateMate stack is rarely a seamless transition due to fundamental differences in DSP block mapping, proprietary IP core lock-in, and differing syntaxes for timing constraints.[20] Relying on the KV260 violates Rule 3 (mirroring vendor SDKs).

The Thermodynamic Calibration Reality Check: Carnot explicitly targets adaptive-K Persistent Contrastive Divergence training utilizing simulated annealing. Physical thermodynamic platforms like the Extropic Z1 operate precisely at the ambient thermal noise limit of their analog circuits.[2] While the architecture includes a programmable "temperature" parameter (β) [1], perfectly calibrating this analog β to match the precise mathematical scheduling required to prevent algorithm divergence in non-convex EBM landscapes is notoriously difficult in physical, mixed-signal silicon. Analog variations across the die, environmental temperature changes, and voltage fluctuations inject unwanted biases, directly violating the strict detailed balance equations mathematically required for theoretically sound Contrastive Divergence. The Carnot core logic will require profound software-based error-correction routines to account for inherent analog drift in any future TSU hardware.

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